

**HISTORICAL-LAYOUT REBUILD — NOT A RELEASED SUBMISSION
VERSION**

**This file uses the current working text for layout checks. The released
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From Configurations to Claims: An Evidence Audit of Urban EV Charging Forecasting

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1 Introduction

Finishing a run is not the same as reproducing the experiment described in a paper. A run can produce a checkpoint and a metric even when the source paper does not fully specify the target, split, preprocessing, or hyperparameters. A well-specified method can also fail for an ordinary engineering reason and produce no comparable result. A single “reproduced/not reproduced” label does not describe these cases well.

This issue matters in urban electric-vehicle (EV) charging forecasting. Public resources cover zonal charging demand in Shenzhen, station occupancy in Paris, workplace charging sessions, and measurements collected across several cities [1–5]. These resources measure different targets. Occupancy rates, port states, session records, charging duration, and energy volume have different denominators, missing-data patterns, and practical meanings. The associated studies also use different model families, including physics-informed graph networks, language-model formulations, exogenous-variable Transformers, linear decompositions, and pretrained forecasters [6–11]. A reported score can therefore be interpreted only when the data meaning, available information, forecast origin, and evaluation protocol are specified.

Forecasting research has long used out-of-sample and rolling-origin evaluation [12, 13]. Later work also describes leakage caused by preprocessing, unsuitable random splits, weak baselines, and metrics that do not match the task [14, 15]. Reproducibility programs encourage researchers to publish code, data, environment details, and reporting artifacts [16]. Large experiment matrices still require three distinctions: incomplete disclosure versus execution failure, an oracle upper bound versus a feasible gain, and a positive result versus a predefined continuation threshold. Without these distinctions, the same evaluation interval may be used repeatedly for tuning.

We audit one UrbanEV configuration inventory and the sequence of model experiments that followed it. We classify configurations into six states, verify target and run identities, and evaluate each proposed extension in a fixed order. The stages are an oracle diagnostic, forecast-origin expert prediction, strict prequential fixed fusion, learned routing, synchronous latency measurement, foundation-model teacher screening, residual distillation, and a Paris development-only teacher test. Metrics remain locked when an artifact is incomplete or invalid. Each model-development stage also uses a time-stamped minimum-effect rule. The study documents how claims were matched to stored artifacts in this project; this procedure has not been validated across separate research teams.

Counting completion and exact reproduction separately gives different totals. Of 1,224 frozen configurations, 865 have completed artifacts, but only 156 meet the joint exact-protocol-and-numerical criterion. In the internal UrbanEV rolling evaluation, strict CAPER–TimeXer fusion reduces macro RMSE by 3.178% relative to the strongest evaluated single model and increases batch-1 median synchronous latency by 48.86% on the frozen laptop-GPU workload. The learned router adds 0.443% over the fixed reference, which is below the predefined 1% progression rule despite a positive block-bootstrap interval. Under the same registered rules, Chronos-2 does not qualify as the project teacher. Residual distillation stops after a 0.119% gain over ground-truth-only

training. Paris student training does not begin because the development-only teacher improves by 0.847% rather than the required 1%. These results apply only to the tested candidates and protocols. They do not evaluate routing, foundation models, or distillation in general.

We organize the study around four research questions:

- RQ1** How many configurations are exactly reproduced, approximately recovered, trend-only, under-identified, failed, or not completed?
- RQ2** How much of the oracle gap remains when CAPER and TimeXer are combined by strict fixed fusion or learned routing?
- RQ3** How do the accuracy results change after including synchronous latency and a predefined minimum-effect rule?
- RQ4** How do artifact checks, control branches, separate data storage, and stopping rules affect which claims are supported by the stored evidence?

The paper reports four main outputs:

- **A six-state configuration table.** The table separates exact reproduction, approximate recovery, trend-only recovery, missing identification, execution failure, and non-completion for 1,224 configurations. It shows why a completed run is not automatically an exact reproduction.
- **An audit checklist and artifact links.** We group records as configurations, predictions, targets, metrics, latency measurements, or protected-storage records. For each group, we state the identity checks, lock conditions, and claim limits used in this study. The ledger links each main claim to its supporting artifact.
- **A fixed sequence of model comparisons.** The experiments use target checks, prequential fitting, control branches, block-resampled uncertainty, and hardware-specific latency. This sequence measures the fixed-fusion gain and shows that later routing, teacher, and distillation results remain below their predefined thresholds.
- **Locked metrics and protected-data controls.** Artifact checks blocked invalid shuffled controls and corrected missing-target representation before metric access. Paris formal and protected target-bearing files were stored separately, so only development targets contributed to the reported Paris result.

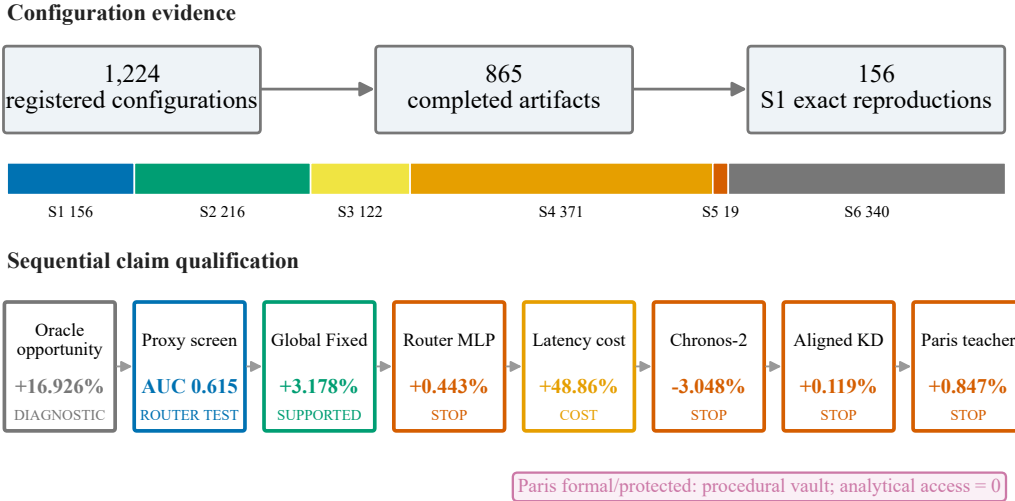


Figure 1: The study reports registration, completed execution, exact reproduction, predictive opportunity, measured engineering cost, and progression decisions separately. A positive diagnostic or comparison does not by itself meet the rule for moving to the next stage. Paris formal and protected targets remain in separate storage with zero analytical use.

Claim boundary. This is an audit and evaluation study, not a new state-of-the-art model paper. Its numerical results concern one UrbanEV inventory, six internal rolling folds at seed 42, one latency platform, and Paris development-only data. UrbanEV and Paris absolute RMSE values are not compared because their panels and targets differ. The 1% threshold is a time-stamped project rule, not a universal definition of scientific or economic importance. We make no claim of protected-test performance, production readiness, universal generalization, or cross-dataset superiority. Paris formal and protected files were moved into separate storage, and that movement is reported as storage rather than analytical access. The conclusions below are limited to the evidence available within this boundary.

Section 2 reviews the relevant datasets and evaluation practices. Section 3 describes the configuration states, data boundaries, metrics, and decision rules. Sections 4–7 report the results and explain what they do and do not support.

2 Related Work

EV charging data and prediction. Public EV charging datasets use different observation units and prediction targets. ACN-Data records workplace charging sessions and supports behavioral and charging-control analyses [4]. UrbanEV provides hourly and finer-resolution zonal charging measurements for Shenzhen, together with prices, weather, points of interest, and spatial descriptors [1]. CHARGED combines charging duration, volume, price, weather, and geospatial fields from six cities [5]. The Paris Smarter Mobility dataset records the states of 273 charging points at 91 stations and supports occupancy forecasting at station, area, and global levels [2, 3]. These datasets support different EV forecasting tasks, but their targets are not directly comparable. A session, an hourly charging-duration total, a zonal occupancy ratio, and a non-available-port rate require different denominators and masks. We therefore check the target used in each experiment and compare absolute errors only within the same protocol.

The target also depends on the intended use. Reviews of open EV load data distinguish session arrivals, connection and charging durations, energy, station load, and occupancy. Other studies consider infrastructure siting, capacity planning, grid operation, and charging schedules [17, 18]. Forecasts may be produced at several aggregation levels or as distributions instead of point estimates. For example, Buzna et al. use information across lower- and higher-level series for hierarchical probabilistic station-load forecasting [19]. These differences mean that the target, aggregation level, forecast type, and horizon must be recorded as part of each protocol. These fields define the valid comparisons in this study. The hierarchical example is background rather than a model contribution of this paper.

Recent EV forecasting studies use several model families. Physics-informed graph learning adds spatial structure and a price–demand prior to regional demand prediction [6]. ChatEV treats heterogeneous charging inputs as a text-to-text prediction problem [7]. TimeXer combines endogenous temporal patches with exogenous variables [8], whereas DLinear provides a simpler decomposition and linear-projection reference for long-horizon forecasting [9]. Cross-paper ranking is outside scope because the tasks, data, splits, and metrics differ. The compact TimeXer and CAPER models in this study are the specific local implementations recorded in our artifacts. Our comparison keeps the target, history, horizon, fold, and evidence source the same.

Time-series evaluation. Prequential evaluation tests each forecast using only information available before its target is observed [12]. Rolling-origin evaluation applies this rule at a sequence of forecast origins and may refit the model between test periods [13]. Cross-validation for dependent data requires care because its validity depends on the prediction setup, stationarity, and residual dependence [20]. Temporally ordered out-of-sample estimators are especially useful under realistic non-stationarity [21]. Our fixed-fusion and routing experiments follow the same rule: labels from the current evaluation fold are not used to fit that fold’s fusion weights or router.

Metric choice also affects the meaning of a reported error. Forecast metrics emphasize different scales, denominators, and target properties, so no single percentage error works for every task [22]. Common problems include missing simple baselines, full-series preprocessing, mismatched horizons, and unsuitable aggregation [14]. Forecasting competitions show the value of careful validation and model combinations, but they also show that complex systems do not always beat simple references [23]. We apply the same checks when deciding whether an experiment should continue. The target must match, the candidate must beat the locked reference, and every registered gate condition must pass.

Reproducibility and evidence auditing. Machine-learning reproducibility programs emphasize code, data, environment details, and structured checklists [16]. Leakage can still cross a nominal train-test boundary through preprocessing, feature construction, temporal misalignment, or repeated analysis [15]. Results may also change with data sampling, initialization, and hyperparameter selection, so small score differences can be dominated by uncontrolled variation [24]. We therefore record run fingerprints, source hashes, and target-array checks. We also label real-target metrics, proxy metrics, oracle diagnostics, latency measurements, and artifact checks separately.

Our stopping rules address changes made after results have been inspected. A model-selection rule can be overfit even when the estimator used inside it is unbiased [25]. Repeated use of a holdout set also weakens its role as fresh evidence [26]. In this project, thresholds were fixed before aggregate access, metrics stayed locked until artifacts closed, invalid controls were rerun as a full set, and later formal and protected periods remained separate. This study documents how the stopping rules were applied and what the resulting records support; it does not introduce a new statistical test.

Ensembling, foundation forecasters, and distillation. Forecast combinations often improve accuracy because the component models make different errors. The M5 competition provides large-scale evidence for both simple and learned combinations [23]. An oracle that selects the better expert after seeing the target is still only an upper bound. It does not show that information available at the forecast origin can recover the same gain. We first measure this oracle bound, then test fixed fusion and learned routing without access to current-fold targets. We also measure synchronous latency to determine the cost of running both experts.

A pretrained forecaster still needs to be tested under the target protocol. Chronos reports zero-shot probabilistic forecasting across varied tasks [10], and Chronos-2 extends this approach to grouped multivariate series and covariates [11]. However, a published aggregate ranking does not determine whether a model is a suitable teacher for a different target, context length, covariate policy, and point-forecast rule. We therefore test one locked Chronos-2 checkpoint under the UrbanEV protocol instead of importing a reported score.

Distillation depends on teacher quality, student capacity, target alignment, and the ground-truth loss. We compare the aligned teacher branch with an identical ground-truth-only student and a shuffled-teacher control. Beating the shuffled control can show that the aligned teacher carries sample-specific information, but it does not establish an improvement over ground-truth training. This comparison follows warnings about selection bias and benchmark variance [24, 25]. The experiment evaluates one fixed student under the registered gate. Cache construction and student expansion stop when the required conditions fail.

3 Methods

3.1 Audit design and record priority

We treated this study as an audit of existing experimental records, not as a search for a new state-of-the-art model. Each result was linked to its protocol, configuration fingerprint, run status, prediction and target identities, metric definition, and allowed claim. When records conflicted,

final decision and summary files checked by a separately instantiated automated artifact audit took precedence over canonical result tables. The tables took precedence over frozen protocol thresholds and historical logs. Completed configurations were reused. Failed or unexecuted configurations could enter later work only under a new versioned identifier, so a later run could not overwrite the earlier status.

The audit separated four questions. *Identifiability* asks whether the disclosed experiment determines a unique protocol and configuration. *Execution* asks whether that configuration reached a terminal run with an artifact. *Predictive evidence* asks whether predictions were evaluated against dataset targets under a time-respecting split. *Engineering qualification* asks whether the effect passed the minimum-effect rule and all consistency checks. Proxy labels were used only to predict which expert would win locally. They were not treated as forecast-accuracy evidence.

Table 1 summarizes the checklist used in this case. For each claim, it records the supporting artifact, identity checks, lock or gate, and strongest allowed wording. The checklist comes from one project and has not been validated as a general method for reducing false discoveries. The Supplementary Material gives the full claim ledger and lock fields.

3.2 Six-level configuration taxonomy

We reclassified a frozen inventory of 1,224 UrbanEV configurations into six mutually exclusive evidence states. The priority order was execution fact (S5 or S6), non-identifiability (S4), and then numerical agreement (S1–S3).

S1: protocol and numerical result identifiable and reproduced. The run covered all six folds, followed the official-compatible protocol, and each of the four disclosed metrics differed by at most 2% from its mapped reference.

S2: protocol broadly consistent, with disclosure differences. The run completed and remained above trend-only evidence, but used an audited correction or had a 2–5% numerical discrepancy.

S3: relative trend only. At least one core metric differed by more than 5%, or the disclosed comparison cell could not be covered completely.

S4: original information insufficient for a unique configuration. The paper omitted an identifying element, such as node identity, or an extended configuration could not be mapped uniquely to a disclosed paper cell.

S5: configuration identified but execution failed. The manifest fixed a configuration, but the registry recorded a failed terminal state.

S6: runnable configuration not actually completed. The manifest froze a runnable configuration that was skipped or never registered as a completed run.

S1–S4 can all have completed artifacts; only S1 denotes exact protocol-and-numerical reproduction under these frozen rules. Classification was deterministic, not an inter-rater coding study. The exact decision rules, priority order, one example per state, and reason fields are given in the Supplementary Material. We did not change the numerical tolerances to alter the counts. The counts describe this UrbanEV inventory and should not be read as a field-wide reproducibility rate.

3.3 Data boundaries

UrbanEV internal rolling boundary. We predicted hourly zonal occupancy rate; other UrbanEV quantities were outside the scope of this audit [1]. Occupied-port counts were divided by the frozen zone capacity. The resulting $4,344 \times 275$ panel contained no non-finite target values and required no target imputation. All same-protocol comparisons used a 168-hour history, horizons $\mathcal{H} = \{3, 6, 9, 12\}$ hours, six canonical expanding-month folds, and seed 42. Each model in a fold-horizon cell used the same target indices, timestamps, target values, and zone ordering. The Supplementary Material lists fold ranges, counts, prediction hashes, and fit sources. This was an internal rolling evaluation, not a blind protected test.

Table 1: Summary of the evidence checks used in this study. The Supplementary Material gives the complete fields and wording limits.

Evidence record	Required checks	Supported statement
Configuration	protocol identity, disclosure mapping, execution state	registered, executed, failed, or S1–S6 evidence state
Prediction/target	index, time, entity, mask, and value hashes	internal/development metric for the named cell only
Diagnostic/proxy	target-dependent or proxy role stated explicitly	opportunity bound or predictability screen, not forecast gain
Engineering measure	device, precision, workload, batch, synchronization	measured latency/throughput on the frozen serving graph
Gate/repair	metric lock, conjunctive decision, parent/child hashes	gate pass/fail or artifact-integrity statement
Protected evidence	role, storage state, access definition and counter	procedural separation and zero analytical use after lock

Procedurally separate Paris boundary. The second boundary used the Paris Smarter Mobility Data Challenge records [2, 3]. For station z and hour t , the development target was the *non-available-port fraction*,

$$y_{t,z} = \frac{C_{t,z} + P_{t,z} + O_{t,z}}{A_{t,z} + C_{t,z} + P_{t,z} + O_{t,z}}, \quad (1)$$

where A, C, P , and O denote Available, Charging, Passive, and Other counts. Charging, Passive, and Other were all counted as unavailable because the target asks whether a port can accept a new user. It does not measure energy demand, session volume, or active charging alone. We locked 50 stations with stable development denominators, totaling 150 ports. The development matrix covered 3,624 hourly timestamps from 3 July through 30 November 2020. Of 181,200 station–hour cells, 180,073 were observed (99.378%).

We divided the remaining timeline into a 50-day formal period (1 December 2020–19 January 2021) and a 50-day protected period (20 January–10 March 2021). Files containing targets for these periods were stored only in a separate vault outside the main project. We checked file identities, row boundaries, and hashes as storage metadata; the analytical-access count remained zero. *Analytical access* means using a formal or protected target for a metric, model selection, feature construction, a statistical summary, or a paper claim. Hashing, moving, and inventorying files count only as storage operations. Two example rows were displayed during pre-lock schema discovery, which we report as a limitation. After the lock, no formal or protected target was used for a metric, model decision, or paper result. The storage migration was recorded, so we do not claim that these bytes were never materialized.

Paris development models shared the same station order, targets, observation mask, history length, and horizons. Within each fold, forward filling was limited to four hours. Remaining missing inputs used a station-by-source-local-hour-of-week median fitted on the training interval, followed by a training-station median. The observation mask was not a model feature, and both training loss and evaluation used observed targets only. The released timestamps contained no UTC offsets, so we ordered the unique timestamps exactly as provided and treated adjacent entries as consecutive data hours. The 24- and 168-hour lags therefore refer to ordinal data hours, not reconstructed UTC hours.

The exact executed model identities, local adaptations, upstream-status limits, and source hashes are provided in the Supplementary Material and in `models/MODEL_SOURCE_MANIFEST.csv`.

3.4 Forecast metrics and prequential fitting

Let $\mathcal{O}_{f,h}$ be the observed target positions for fold f and horizon h , including all valid timestamps and spatial units. For method m , cell RMSE was

$$R_{f,h}^{(m)} = \sqrt{\frac{1}{|\mathcal{O}_{f,h}|} \sum_{(t,z) \in \mathcal{O}_{f,h}} \left(\hat{y}_{f,h,t,z}^{(m)} - y_{f,h,t,z} \right)^2}. \quad (2)$$

The primary endpoint was the equal-weight mean of cell RMSEs,

$$\bar{R}^{(m)} = \frac{1}{|\mathcal{F}||\mathcal{H}|} \sum_{f \in \mathcal{F}} \sum_{h \in \mathcal{H}} R_{f,h}^{(m)}. \quad (3)$$

This average gives every fold–horizon cell equal weight, so cells with more observations do not dominate the main result. MAE, WAPE, sMAPE, and RAE were retained as secondary diagnostics. For a candidate c and frozen reference r , positive relative gain denotes lower RMSE:

$$G(c; r) = 100 \frac{\bar{R}^{(r)} - \bar{R}^{(c)}}{\bar{R}^{(r)}}. \quad (4)$$

Absolute UrbanEV and Paris RMSE values were not compared because the datasets differ in spatial unit, target construction, missingness, and fold definition.

We used strict prequential fitting [12, 13]. A parameter evaluated in fold f could use only validation data that preceded that fold or out-of-fold predictions from earlier folds. For the two UrbanEV experts, CAPER (C) and compact TimeXer (T), a fixed forecast was

$$\hat{y}_{f,h}^{\text{fixed}} = \alpha_{f,h} \hat{y}_{f,h}^T + (1 - \alpha_{f,h}) \hat{y}_{f,h}^C, \quad 0 \leq \alpha_{f,h} \leq 1. \quad (5)$$

The Global Fixed variant shared one α_f across horizons, whereas the Horizon Fixed control used four $\alpha_{f,h}$. Fold 1 used only pre-test validation predictions; folds 2–6 used only earlier-fold out-of-fold predictions. The target-dependent oracle selected the expert with smaller pointwise error. It was an upper bound that cannot be used in practice because it observes the target.

The expert-win classifier used 21 forecast-origin features whose timestamps did not exceed the forecast origin. Six described the experts: CAPER prediction, TimeXer prediction, signed and absolute disagreement, expert mean, and normalized horizon. Fifteen described the observed state: current occupancy, one-step change and its magnitude, daily and weekly deviations and their magnitudes, 24-hour mean, standard deviation and turnover, normalized capacity, and sine/cosine encodings of hour and weekday.

The AUC and Brier score measured how well the classifier predicted which expert would have lower error. They did not measure forecast accuracy or a causal mechanism. We evaluated learned routing separately with real target RMSE. The routing methods were a hard gate, a raw-probability soft gate, and a direct-loss multilayer perceptron. The locked router protocol named the direct-loss MLP as the primary comparison before aggregate Gate 4 access, even though the later raw-soft point estimate was lower.

3.5 Uncertainty estimates and progression rules

Paired uncertainty estimates resampled target timestamps in contiguous 24-hour blocks and kept all spatial units at a timestamp together. The router used 1,000 bootstrap replicates. Distillation and Paris teacher qualification used 5,000 replicates.

For Paris, we first found the timestamps shared by all four horizons in a fold and divided the ordered sequence into consecutive blocks. Each replicate drew the original number of blocks with replacement and used the same sampled timestamps for all four horizons. We then recomputed every cell RMSE, selected the better single-model reference inside the replicate, and calculated

equal-cell macro gain. Each monthly sequence ended with a 12-hour partial block. Keeping this block made replicate length variable, but every original timestamp had the same expected inclusion. We did not test 48- or 168-hour alternatives after seeing the result. We report protocol-specific paired 95% bootstrap intervals. They are descriptive, and a positive interval could not override a failed minimum-effect rule.

Every listed condition had to pass. The 1% rule was fixed before aggregate access as a conservative project-level threshold for authorizing additional model search, cache construction, and added serving complexity. It was not estimated from the evaluated outcomes. It was also not derived from charger-operator utility and is not a universal significance threshold.

The lock artifacts predate their metric unlocks. Their complete hashes and freeze/unlock times are retained in the public machine-readable audit receipt. The displayed prefixes are router protocol `0b1fc56d...e7f18d35`, distillation protocol `fcd07f18...dfe383bc`, and Paris gate specification `96babeb1...90e9561`.

The UrbanEV router had to beat TimeXer and both fixed controls and improve macro RMSE by at least 1%. It also had to win at least five of six folds and 18 of 24 cells, have a paired interval lower bound above zero, and avoid more than 1% deterioration at any horizon. Chronos-2 teacher qualification used the same 1%, fold, cell, and horizon criteria. It also required matching target identity, causal input exposure, revision reproducibility, and cache feasibility.

Residual distillation compared the same 338-parameter student trained with ground truth only, aligned teacher residuals, or a 24-hour-block shuffled teacher. Aligned KD had to improve over GT-only by at least 1%, win five of six folds and 18 of 24 cells, have a positive paired lower bound, and keep every horizon within 1%. It also had to beat the shuffled control in macro RMSE, at least five folds, and the paired interval.

Paris teacher qualification was development-only. The strict prequential CAPER–TimeXer teacher had to improve over the replicate-wise best single model by at least 1% and win at least three of four folds and 12 of 16 fold–horizon cells. It also needed a paired lower bound above zero, no more than 1% deterioration at any horizon, and matching target, station, mask, timestamp, and origin identities. A value below 1.000% failed the point-effect condition. We did not round the value or revise the threshold.

3.6 Teacher, distillation, and latency settings

We screened the official `amazon/chronos-2` model [11] from a locked local snapshot at commit prefix `29ec3766`; the complete revision is retained in the model-lock artifact. The screen used `chronos-forecasting 2.2.2`, zero-shot FP32 inference, a 168-hour multivariate context, no future covariates, and the native median (Q0.5) forecast. Predictions were clipped to $[0, 1]$ for the frozen primary UrbanEV RMSE, matching the existing occupancy-rate evaluation.

The residual-distillation teacher was the audited Global Fixed fusion. The student combined a CAPER anchor with a shared DLinear residual branch [9]. GT-only, aligned-KD, and shuffled-teacher branches shared architecture, initialization, optimizer, epochs, data, and seed; only the teacher-residual term differed. This negative control tested whether any benefit required sample-aligned teacher information rather than merely an additional regularizer.

Synchronous latency was measured on one RTX 4060 Laptop GPU in FP32 with a sequential single-device setup. One request produced all four horizons for all 275 UrbanEV zones. Each method–batch combination used 50 warm-up iterations and 1,000 timed iterations in each of five separately launched processes. Batch sizes were 1, 8, and 32 origins.

Model-only time used CUDA events and synchronized the final event. End-to-end wall-clock time synchronized the GPU before stopping. It included input preparation, transfers, four-horizon inference, fusion, and output transfer, but excluded model loading and checkpoint I/O. The measurement therefore represents a warm resident service, not cold start. Batch-1 end-to-end median latency was the primary engineering endpoint. The Supplementary Material reports the full batch 1/8/32 table and the software and hardware details.

3.7 Fail-closed artifact audit

The pipeline kept evaluation metrics locked until every registered run closed and its predictions passed identity checks. An invalid control mapping, target mismatch, boundary violation, incomplete provenance, or representation error blocked metric access.

Repairs were non-selective. If one control family was defective, the full family was rerun rather than only the cells where an error was found. Each repair receipt recorded the old and new hashes and whether predictions changed.

A separately instantiated automated artifact audit is a read-only audit created outside the workflow that generated the artifact. It is not an external human replication, blinded coding exercise, or multi-laboratory validation. The audit could block execution or metric access, but it could not edit predictions or lower thresholds. Decision files were retained with negative results. The Supplementary Material lists the states, lock fields, and unlock conditions. The audit still relied on project-generated receipts, which remains a limitation.

The Paris storage guard also separated file storage from analytical use. Formal and protected target-bearing files remained in vault-only storage, the main project contained no matching target source, and the analytical-access count remained zero.

4 Results

4.1 Many configurations completed, but only 156 met S1

The frozen registry contained 1,224 unique configurations with no duplicate run identifier or configuration fingerprint. Of these, 865 (70.67%) had completed artifacts, 19 had failed terminal states, and 340 had not been executed. Completion did not imply exact reproduction: only 156 configurations (12.75%) satisfied S1, while 216 (17.65%) were S2, 122 (9.97%) were trend-only S3, and 371 (30.31%) were non-identifiable S4. Thus, the S1 count was less than one fifth of the completed-artifact count.

The distribution also depended on the original UrbanEV source table. UrbanEV source Table 3 contained all 156 S1 configurations, alongside 180 S2, 94 S3, six S5, and 44 S6 configurations. In contrast, 215 of 216 source Table 4 configurations were S4 under the frozen mapping rule. Source Table 5 contained 156 S4 and 296 S6 configurations among 528 entries. These patterns identify where disclosure and execution evidence was limited; they do not rank the predictive quality of the models appearing in those source tables.

4.2 Fixed fusion recovered part of the oracle gain

The baseline experiment completed all 72 registered runs (three models, six folds, and four horizons). Among the evaluated single models, compact TimeXer had the lowest macro RMSE at 0.073709. DLinear reached 0.075362, NLinear 0.075711, and GBDT-Forecaster 0.078165; CAPER’s existing same-target result was 0.075507. TimeXer was best only among the models tested under this UrbanEV protocol.

CAPER and TimeXer still made different local errors, leaving a sizeable oracle gap. Their mean residual Pearson and Spearman correlations were 0.8422 and 0.7638, respectively. A target-dependent oracle reduced macro RMSE to 0.061233, a 16.926% improvement over TimeXer. This was only an upper bound because it used the realized target to choose the expert.

Global Fixed recovered part of this oracle improvement. Its macro RMSE was 0.071367, 3.178% below TimeXer. It improved all six fold means and all 24 fold-horizon cells, with horizon-level gains from 2.890% to 3.645%. A paired block interval was not part of this earlier frozen accuracy protocol, so the result was not treated as passing the later router or teacher gate. The Horizon Fixed control was nearly identical at 0.071372. Post-hoc horizon-specific weights were therefore not needed for the fixed-fusion point result.

Table 2: Six-level evidence taxonomy in the frozen UrbanEV configuration inventory. Percentages use all 1,224 registered configurations as the denominator.

Code	Evidence status	Count	Share
S1	Protocol and numerical result identifiable and reproduced	156	12.75%
S2	Protocol broadly consistent; disclosure differences remain	216	17.65%
S3	Relative trend only	122	9.97%
S4	Original information cannot identify a unique configuration	371	30.31%
S5	Configuration identified; execution failed	19	1.55%
S6	Runnable configuration not actually completed	340	27.78%

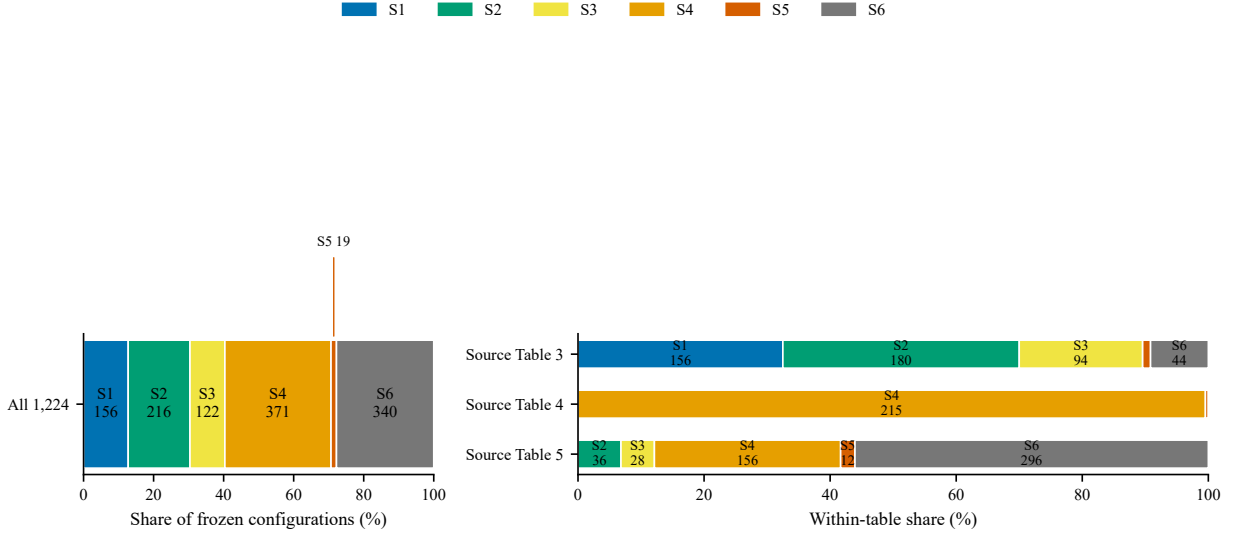


Figure 2: Evidence status in the frozen UrbanEV inventory, overall and by source table. Completed execution and exact reproduction are separate quantities; status counts do not measure intrinsic model quality.

Predicting which expert would win was harder than the oracle gap suggested. A rolling classifier using expert outputs and disagreement alone had mean AUC 0.5370 (range 0.5227–0.5701). Adding 21 forecast-origin features increased LightGBM mean AUC to 0.6151. The minimum and maximum fold AUC were 0.5660 and 0.6682, and the mean Brier score was 0.2388. These proxy results supported running one small router experiment. They were not evidence of better forecasts.

4.3 The learned router produced a positive but sub-threshold gain

The hard router underperformed Global Fixed, with macro RMSE 0.072759. Raw-probability soft routing obtained the lowest observed router RMSE, 0.070979, corresponding to a 0.543% gain over Global Fixed. The locked primary direct-loss MLP obtained RMSE 0.071050, a 0.443% gain. The better secondary point estimate did not replace that primary comparison. The MLP won five of six folds and 18 of 24 cells, improved every horizon, and had a paired 24-hour-block gain interval of [0.275%, 0.634%]. It captured only 3.123% of the oracle gap.

Although the consistency checks passed, the 0.443% gain remained below the frozen 1% threshold. Gate 4 therefore failed, and router development stopped. The result does not show that routing had no signal. It shows that the measured gain was too small to continue under the registered rule.

Table 3: Same-protocol UrbanEV accuracy results over six rolling folds and four horizons (24 equal-weight cells, seed 42). The oracle uses target-dependent expert choice and is not deployable.

Method	Cells	Macro RMSE	Macro MAE	Evidence role
GBDT-Forecaster	24	0.078165	0.049388	lightweight baseline
NLinear	24	0.075711	0.047894	lightweight baseline
CAPER	24	0.075507	0.048480	expert
DLinear	24	0.075362	0.048041	lightweight baseline
TimeXer	24	0.073709	0.046367	best evaluated single model
Global Fixed	24	0.071367	0.044949	strict prequential fixed fusion
Horizon Fixed	24	0.071372	0.044951	horizon-specific fixed control
Oracle	24	0.061233	0.035023	target-dependent upper-bound diagnostic

4.4 Global Fixed improved RMSE but increased latency

The synchronous benchmark completed 90 of 90 process jobs and retained 90,000 raw timing rows. At batch one, TimeXer required 14.067 ms median and 15.952 ms P95 end-to-end latency. Global Fixed required 20.940 ms median and 33.123 ms P95. Thus, the 3.178% internal RMSE improvement came with 48.86% higher P50 latency. P95 latency was also higher. NLinear and DLinear were much faster (0.674 and 1.166 ms P50) but had higher RMSE. In this single-GPU sequential workload, Global Fixed had the lowest RMSE, but its latency was substantially higher. Batch 8/32, model-only, throughput, environment, and portability details appear in the Supplementary Material.

4.5 Chronos-2 did not qualify as the UrbanEV teacher

The locked Chronos-2 screen completed all 24 fold-horizon cells and 5,960 forecast origins under the zero-shot, no-future-covariate, median point-forecast setting. Target, index, timestamp, and zone identities passed. All inputs ended no later than the forecast origin, and a repeated sentinel prediction had maximum absolute difference zero.

Chronos-2 macro RMSE was 0.073542, compared with 0.071367 for Global Fixed, a relative gain of -3.048% . It beat the control in one of six folds and four of 24 cells. Its RMSE deteriorated by 2.138% – 3.932% across the four horizons. Chronos-2 had lower secondary MAE (0.042250 versus 0.044949), but the registered decision rule used RMSE as the primary endpoint and MAE only as a secondary diagnostic. Chronos-2 therefore failed this teacher screen, and Global Fixed remained the teacher for this project. This result is not a general comparison between foundation forecasters and local models.

4.6 Aligned distillation did not improve consistently over GT-only

The final distillation matrix contained 72 audited cells: GT-only, aligned-KD, and shuffled-teacher branches across six folds and four horizons. Macro RMSE was 0.075568 for GT-only, 0.075478 for aligned KD, and 0.075674 for the shuffled control. Relative to GT-only, aligned KD improved RMSE by 0.119%, won four of six folds and 14 of 24 cells, and had a paired interval of $[-0.064\%, 0.303\%]$. It failed the frozen gain, fold, cell, and positive-lower-bound criteria, while horizon protection passed.

Aligned KD did outperform shuffled-teacher training by 0.259%, with five of six fold wins and a paired interval of $[0.151\%, 0.367\%]$. This result is consistent with some sample-aligned teacher information, but it did not produce a stable improvement over the identical GT-only student. The D1 gate failed, `distill_v1` terminated, and neither student latency nor multi-seed or protected evaluation was run.

Table 4: Batch-1 end-to-end accuracy–latency trade-off on the frozen RTX 4060 Laptop GPU, FP32 single-device sequential workload.

Method	RMSE	P50 ms	P95 ms	Origins/s
NLinear	0.075711	0.674	1.056	1483.7
DLinear	0.075362	1.166	1.763	857.9
GBDT-Forecaster	0.078165	6.242	7.195	160.2
CAPER	0.075507	7.934	10.798	126.0
TimeXer	0.073709	14.067	15.952	71.1
Global Fixed CAPER+TimeXer	0.071367	20.940	33.123	47.8

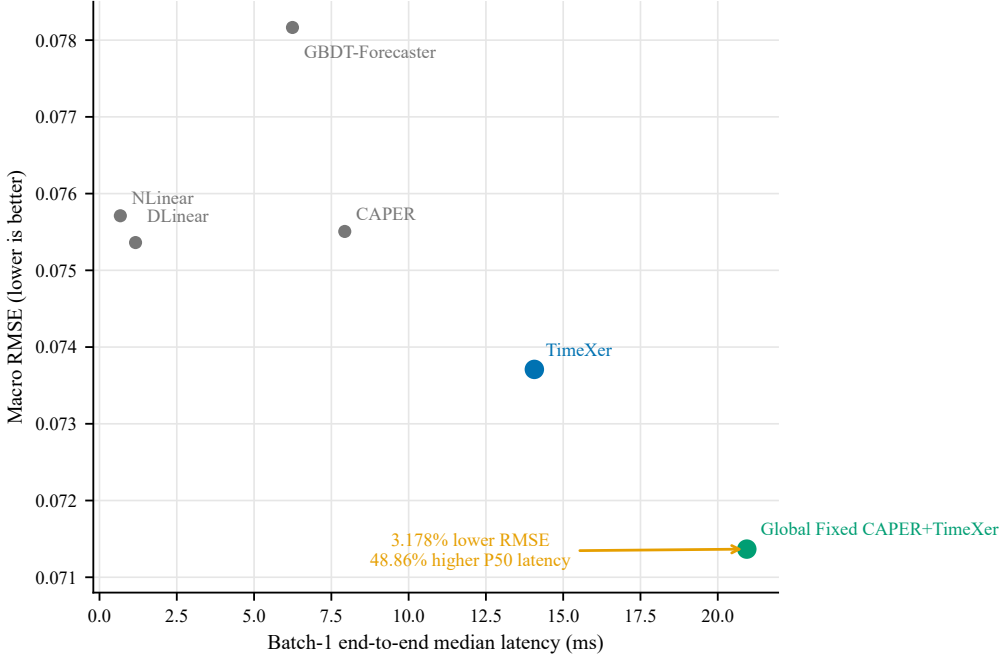


Figure 3: Batch-1 UrbanEV accuracy–latency frontier on the frozen RTX 4060 Laptop GPU in FP32. The arrow from TimeXer to Global Fixed represents 3.178% lower RMSE and 48.86% higher median end-to-end latency.

4.7 Paris gains were consistent but remained below the 1% rule

Paris non-available-port-fraction results are reported only within the development boundary and are not compared numerically with UrbanEV. They are not energy-demand or session-volume forecasts. Among simple development baselines, last observation had macro RMSE 0.302630, daily lag 0.331654, and weekly lag 0.381137. Across the 16 fold–horizon cells, the locally audited Paris CAPER adaptation obtained RMSE 0.244601 and the locally audited TimeXer adaptation 0.242542. The strict prequential fixed teacher reduced RMSE to 0.240488; the target-dependent oracle diagnostic was 0.217661.

The teacher improved over TimeXer by 0.8469%, won all four folds and all 16 cells, and had a paired interval of [0.6867%, 1.0088%]. Every bootstrap replicate selected TimeXer as the best single reference, and no horizon deteriorated. The result was consistent, but the point estimate remained below the frozen 1% progression rule. Because every condition was required, the teacher gate failed. Student training stopped, the value was not rounded into a pass, and no formal or protected access was authorized.

Formal- and protected-role target-bearing sources existed only in the procedurally isolated evidence vault. The storage boundary was materialized and hash-audited, but analytical access count remained zero; no post-lock target value or metric from either role contributed to the Paris

result or this paper.

4.8 Artifact checks blocked metric access until errors were corrected

The first M7-C matrix audit passed 69 of 72 cells and found that three shuffled-control mappings left the data unchanged. Metrics remained locked. The pipeline invalidated and reran all 24 shuffled controls instead of rerunning only the three detected cells. The final audit passed all 72 cells before D1 metrics were computed.

The Paris pipeline used the same fail-closed rule. The first automated artifact audit blocked the 32 development runs because eight implementation or boundary requirements were unresolved. After those issues were corrected, a final automated differential audit issued the internal status `DEPLOY_PASS`. This status authorized the development runs only; it did not indicate deployment readiness. All 32 model jobs then closed before metric access.

A representation audit later found fill values at masked target positions in 24 of 32 artifacts, covering 10,236 entries. The repair restored missing-value representation without retraining and left every prediction unchanged. Metrics had still not been accessed. A subsequent prediction audit passed 32 of 32 artifacts and then unlocked the development metrics. These automated audits were not external human replications.

The reported negative decisions were computed only after the registered artifact checks passed. These checks do not show that auditing improved forecast accuracy. Throughout the Paris stage, formal and protected sources remained in vault-only storage and the analytical-access count stayed at zero.

The Supplementary Material links each research question to its result and supporting artifact.

Table 5: Stage-specific qualification outcomes. Gains use different frozen references and are not absolute cross-stage or cross-dataset comparisons. “–” for Chronos-2 means that its frozen screen did not register a paired interval, not zero uncertainty.

Stage	Frozen reference	Gain	Paired 95% CI	Primary point gate	Decision
Router MLP	Global Fixed	+0.443%	[0.275, 0.634]%	1.0%	FAIL
Chronos-2	Global Fixed	-3.048%	–	1.0%	FAIL
Aligned KD	GT-only student	+0.119%	[-0.064, 0.303]%	1.0%	FAIL
KD vs shuffled	shuffled teacher	+0.259%	[0.151, 0.367]%	positive control contrast	PASS SUBCHECK
Paris teacher	Paris TimeXer	+0.847%	[0.687, 1.009]%	1.0%	FAIL

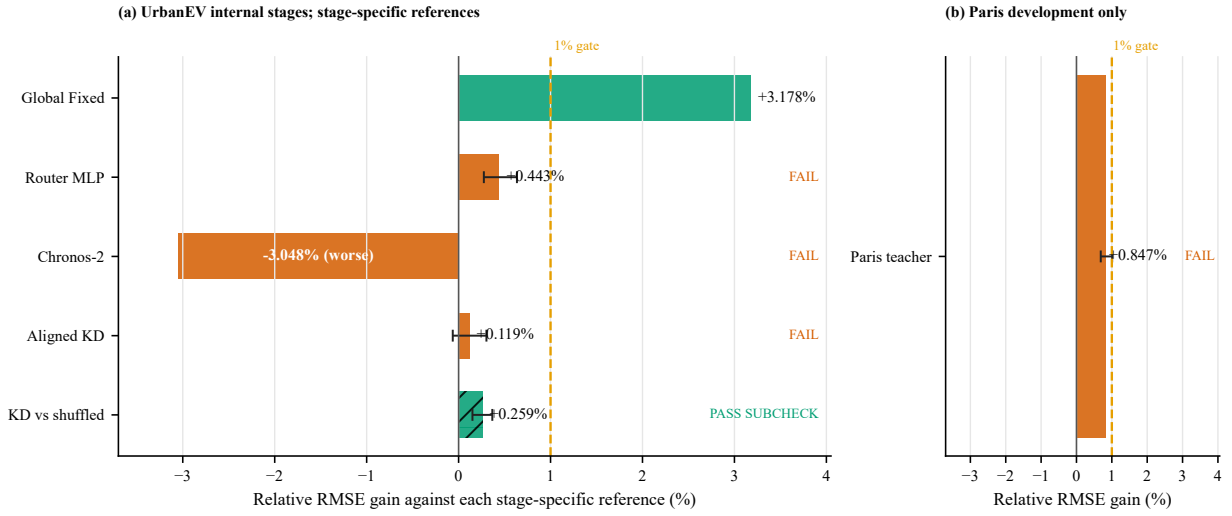


Figure 4: Stage-specific relative gains against each frozen reference. UrbanEV and Paris use different panels, targets, and references; the plot compares gate outcomes rather than absolute RMSE. The dashed 1% line is the project-specific minimum effect, not a universal utility threshold.

Table 6: Fail-closed interventions performed before scientific interpretation. Repairs did not change model predictions or optimize observed metrics.

Event	Detection	Required action	Final state
M7-C control	3 identity mappings detected (69/72 initial audit pass)	all 24 shuffled controls rerun	72/72 final audit PASS
Paris predeployment review	automated artifact audit: BLOCK	8 blockers repaired; DEPLOY PASS	pre-run authorization only; metrics locked
Paris target repair	24 runs; 10,236 masked targets restored	restore missing representation; predictions unchanged	metrics still locked
Paris metric unlock	32/32 jobs closed	prediction identity audit	prediction audit PASS; metrics unlocked; teacher gate FAIL; formal/protected analysis locked; analytical access 0

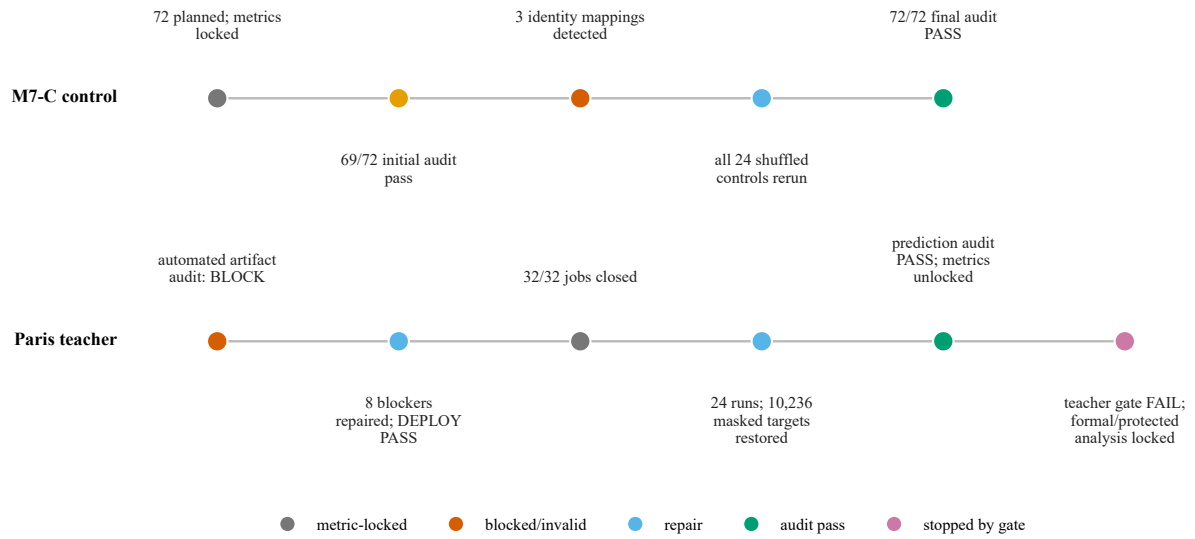


Figure 5: Fail-closed audit timelines. Invalid controls and missing-target representation were corrected before metric interpretation; the corrections did not alter model predictions or authorize formal/protected analysis.

5 Discussion

5.1 Completion does not equal reproduction

The configuration audit shows a difference that a completed-run count would miss. Of the 1,224 registered UrbanEV configurations, 865 have completed artifacts, but only 156 meet S1. S1 requires an identifiable protocol and numerical agreement within the prespecified tolerance. Other completed artifacts fall into different categories. Some broadly match the disclosure (S2), some recover only relative trends (S3), and some cannot be mapped to one source configuration (S4). A successful process exit shows that an implementation ran. It does not show that the implementation matches the reported experiment.

The status categories also should not be read as model-quality scores. For example, many S4 entries occur in one source table because identifying information is missing. That concentration does not show that the associated models are weak.

Using six categories retains more information than a binary “reproduced/not reproduced” label. It keeps partial results separate from exact reproduction and follows prior work that distinguishes artifacts, procedures, and numerical outcomes [16, 24]. In this inventory, the difference is visible in the reported rates: 70.67% of configurations have completed artifacts, whereas 12.75% meet S1. These percentages answer different questions.

5.2 The oracle gap was much larger than the feasible router gain

CAPER and TimeXer made different local errors. A target-dependent oracle that selected the lower-error expert for each target reduced RMSE by 16.926% relative to the best evaluated single model. The oracle is useful as an upper bound, but it observes the target and cannot be used at forecast time. When selection was restricted to information available by the forecast origin, the improvement was much smaller. Global Fixed retained a 3.178% internal RMSE improvement, while the learned direct-loss router added only 0.443% over Global Fixed and captured 3.123% of the oracle gap.

The two values use different information sets. The first number uses errors observed after the outcome, whereas the second uses only information available before the outcome. The state-feature classifier’s mean AUC of 0.6151 shows moderate ability to predict which expert will have lower error. It does not show causal knowledge of charging demand or an equivalent forecast gain. For this reason, we report the oracle, proxy classification, forecast results, and engineering measurements as separate analyses.

The latency measurements add another constraint. On the frozen RTX 4060 Laptop GPU, batch-one end-to-end P50 latency increased from 14.067 ms for TimeXer to 20.940 ms for Global Fixed, a 48.86% overhead. P95 increased from 15.952 to 33.123 ms. Thus, the result supports an internal accuracy gain with a measured cost on this hardware setup. It does not support a general deployment recommendation.

5.3 A positive confidence interval does not satisfy the 1% rule

Several experiments produced positive signals but did not meet their predefined rules. The router improved on Global Fixed by 0.443%, with a paired 24-hour-block interval of [0.275%, 0.634%]. The interval is above zero but below the frozen 1% threshold. The Paris teacher won all four development folds and all 16 fold-horizon cells. Its interval was [0.687%, 1.009%], but its point improvement was 0.847%. The rule required a point improvement of at least 1%, so the teacher did not qualify. The aligned distillation branch was weaker: its 0.119% improvement over the identical GT-only student had an interval of [−0.064%, 0.303%] and failed the fold and cell requirements.

These decisions do not mean that effects below 1% are scientifically meaningless. The threshold is a local rule for continuing engineering work, not an operator utility model or a universal statistical constant. The interval describes uncertainty in the signed effect under the registered

bootstrap procedure. The progression rule asks whether the effect is large and consistent enough to justify another stage. Using the interval in place of the threshold would change the decision rule after seeing the result.

For context only, a 0.5% rule would still reject the router, Chronos-2, and aligned KD, but would change the Paris development decision. A 2% rule would reject every downstream candidate. These calculations do not change the frozen 1% decisions.

5.4 Stopping rules limited repeated search on the same data

The stopping rules prevented small positive results from starting open-ended searches on the same evidence interval. Router v1 stopped after its effect-size failure. Chronos-2 remained a recorded zero-shot teacher screen after its -3.048% gain against Global Fixed. Distillation v1 stopped after the aligned branch failed its main rule, even though it beat the shuffled control by 0.259% . The Paris teacher also stopped at development; formal and protected targets were not opened after the near-threshold result. Each negative result therefore corresponds to a prespecified comparison rather than one step in an unreported adaptive search [25, 26].

The artifact checks had a similar role. In the distillation matrix, three identity mappings invalidated the shuffled controls while metrics remained locked. All 24 shuffled-control cells were rerun, and metrics were unlocked only after the 72/72 audit passed. In Paris, the first automated predeployment audit blocked execution on eight issues. After correction, 32 prediction artifacts closed without evaluation metrics. A later representation audit changed 10,236 masked target entries back to missing values across 24 runs without changing predictions. Metrics remained locked until all 32 artifacts passed the identity audit. These repairs ensure that the reported metrics use the intended artifacts. They do not show that repair improved model accuracy.

5.5 The vault records data separation, not model performance

Paris formal and protected target-bearing files are stored outside the main project. The inventory contains 48 non-self-referential files, including 29 hidden Git files and the object pack that could reconstruct removed target sources. Every entry matches its recorded byte count and SHA-256 hash. The formal and protected shards cover 2020-12-01 through 2021-01-19 and 2021-01-20 through 2021-03-10. The files were moved once, but the protected analytical-access count remains zero. Failed partial files are also inventoried and were not analyzed.

This record supports one limited claim: no post-lock Paris formal or protected target contributed to model selection, metric computation, or a paper result. It does not mean that the bytes never existed, that two pre-lock schema rows were never visible, or that Paris development is a blind test. The controls address accidental analysis, stale-cache reuse, path leakage, and recoverable Git history. They do not protect against a malicious authenticated user or an operating-system compromise, and the vault is not encrypted storage. Its purpose is to make the zero-access statement checkable and to retain the targets for a later versioned protocol.

5.6 Practical reporting suggestions

Based on this case, authors should state whether a target measures energy, sessions, active charging, occupancy, or port non-availability. They should also report denominators and observation masks, keep model-search periods separate from later operational evaluation, and define the request shape used for latency. Reviewers can then check whether a completed run is still underidentified, whether an oracle or proxy is being reported as forecast performance, and whether a positive interval is being used in place of a predefined minimum-effect rule.

Operational studies may also need measures beyond macro RMSE, such as peak-period error, high-occupancy recall, calibrated intervals, or denied-service proxies. This paper does not estimate those quantities. The Supplementary Material contains an author/reviewer checklist derived from this case.

6 Limitations

Scope of the registry. The 1,224-configuration taxonomy describes one frozen UrbanEV reproduction inventory. Its S1–S6 assignments depend on the available source disclosure, deterministic mapping rules, and internally fixed 2% and 5% numerical tolerances. The classifications were not independently double-coded, and no inter-rater reliability estimate is available. The resulting 12.75% S1 rate is not an estimate of reproducibility across urban forecasting as a field. Status counts also measure evidence identifiability and execution state, not intrinsic model quality.

Validation outside this project. The audit procedure comes from one self-audited project. Separately instantiated automated artifact audits checked specific protocol and result packages, but they were not external human replications. No external team has applied the full wording rules, audit procedure, or S1–S6 decision tree to another benchmark. The paper therefore reports one procedure and one worked example. It does not show that the procedure improves decisions or reduces false discoveries in other groups.

Internal and development-only evaluation. The UrbanEV comparisons use six internal rolling folds, four horizons, and seed 42. Paris teacher qualification uses four development folds over a different panel, non-available-port target, missingness mask, and local audited model adaptations. Absolute UrbanEV and Paris RMSE values are therefore not comparable. The windows also differ behaviorally: Shenzhen covers September 2022–February 2023, whereas Paris development covers July–November 2020, when pandemic-era mobility, policy, infrastructure, pricing, land use, and fleet composition could differ. These checks address protocol leakage but not the listed transport-domain differences. No UrbanEV protected evaluation was opened, and no post-lock Paris formal or protected target was used analytically. The results do not establish blind-test generalization, robustness across seeds or cities, production readiness, or state of the art.

Restricted candidate and implementation set. The baseline conclusion applies only to the evaluated candidates. The Chronos-2 screen fixes one official model revision, zero-shot FP32 inference, and no future covariates; it does not establish that foundation models generally underperform on charging data. Likewise, the CAPER and compact TimeXer Paris implementations are local audited adaptations rather than claims about every upstream implementation. The residual-distillation result concerns one 338-parameter student and one seed. The fixed protocol did not include a student architecture search, multi-seed expansion, or latency evaluation.

Oracle, proxies, and uncertainty. The target-dependent oracle gives only an upper bound on expert complementarity. The expert-advantage classification task uses a proxy label derived from relative expert errors; its AUC does not prove causal predictability or forecast improvement. Paired block-bootstrap intervals account for temporal dependence under the chosen 24-hour blocking scheme, but they do not remove all uncertainty from fold construction, model selection, or the small number of seeds. Each Paris fold retained a trailing 12-hour partial block, and no post-hoc 48- or 168-hour sensitivity was run. Its interval should therefore be read under that exact resampling procedure. The 1% minimum-effect rule has no accompanying economic utility model and should not be generalized as a universal significance threshold.

Hardware-specific latency. Latency was measured in FP32 under single-device sequential execution on one RTX 4060 Laptop GPU; batch one was the frozen primary endpoint, while batches 8 and 32 are descriptive appendices. A request produced predictions for 275 zones and four horizons. Kernel fusion, parallel expert execution, quantization, compilation, batching, cold start, a different accelerator, or a serving system could change both the absolute timings and

the fusion overhead. The measured 48.86% P50 overhead applies to the frozen warm-resident workload and should not be assumed to hold for other deployment settings.

Audit coverage. Fail-closed checks detected identity shuffles, target-representation errors, incomplete provenance, and premature metric access. These checks reduce known integrity risks but cannot prove the absence of every implementation defect. The repairs were verified through hashes, identities, and unchanged predictions; they were not independently reimplemented from scratch.

The automated Paris result audit retained three non-decisive warnings. The teacher-weight CSV lacked complete upstream provenance, the pre-unlock state lacked a separate hash snapshot, and the receipt-update chain was weaker than the prediction-identity audit. Because the point estimate is below the frozen gate, these warnings do not change the recorded decision, but they limit stronger claims about the audit process.

Two example source rows were displayed during early schema discovery before the Paris boundary lock, when no model or selection rule existed. This event was logged and was not used for model choice, but it prevents an absolute claim that no target value was ever visible before boundary creation.

Dataset-license metadata. The Paris challenge repository and the associated Zenodo record expose different license labels (ODbL and CC BY 4.0, respectively) [2, 3]. This paper treats them as source-specific metadata and does not resolve the legal relationship between the records. Downstream redistribution should therefore verify the applicable terms at the exact source used.

Vault boundary. The final vault manifest covers 48 non-self-referential files, including 29 Git files, and all recorded hashes verify. Analytical access to protected labels is zero, but storage migration materialized formal and protected shards and retained redundant raw, reference, final-shard, and failed-partial copies. The partials were inventoried but not analyzed. The vault inherits filesystem permissions, including modify rights for authenticated users; it is procedural isolation rather than encryption or hardened authorization. Some tutorial figures and score summaries were generated from targets and remain outside the isolated file set, but they cannot reconstruct station-level labels. These properties limit the claim to verified source separation and zero analytical use.

Negative-result interpretation. A failed frozen gate does not prove that an entire method family cannot work. It shows only that the tested candidate did not meet the registered rule on this reference and data interval. Likewise, a passed subcheck—such as aligned distillation outperforming a shuffled teacher—does not change the failed main comparison. Our conclusions describe what these experiments support, not whether the method family can ever succeed.

7 Conclusion

This study considered four questions separately: whether a configuration is identifiable, whether an implementation completed, whether it carries predictive signal, and whether its gain meets the rule for moving to the next stage. In the UrbanEV registry, 865 of 1,224 configurations have completed artifacts, but only 156 meet S1. Completion is useful information, but it is not the same as exact reproduction.

The later model experiments produced different results at each stage. The target-dependent oracle reduced RMSE by 16.926% relative to TimeXer. Strict prequential fixed fusion retained a 3.178% internal gain and increased batch-one P50 latency by 48.86% on the frozen device. Learned routing, the Chronos-2 teacher screen, aligned residual distillation, and the Paris development

teacher were each evaluated under their registered rules, but none met its progression rule. Positive router and Paris intervals did not replace the project-specific 1% point requirement.

Metric locks, full control reruns, prediction-identity checks, and time-stamped stopping rules made each negative result traceable to a prespecified comparison. Paris formal and protected targets were kept out of the present analysis. The separate store contains 48 non-self-referential files, including 29 recoverable Git files, and the protected analytical-access count remains zero. This is a documented storage rule; it is not encrypted storage or a protected-test result.

The main result concerns evaluation practice rather than a new model architecture. In this case, separate checks make clear which conclusions the stored evidence supports. The wording rules and audit checklist have not yet been tested by another team or on another benchmark. Future work should apply a newly frozen model and loss specification to an untouched data boundary, use multiple seeds and hardware environments, and test the same stopping rules independently. At present, the evidence supports an internal accuracy–cost result and the recorded negative decisions, while formal and protected targets remain unused for later evaluation. It does not support a state-of-the-art or deployment-readiness claim.

Archive-only audit tables

Table 7: Fail-closed audit steps used in this study. Each transition requires a checkable artifact; a failed condition moves to *blocked* rather than metric interpretation.

Stage	Required state	Minimum record	Fail-closed transition	Supported statement
1. Register	protocol fixed before aggregate evidence	data role, target, split, forecast origin, model/config ID, seed, endpoint, controls, gate, stop rule	missing or mutable field → BLOCKED	registered only
2. Execute	registered configuration	run ID, configuration fingerprint, code/environment identity, attempt state, immutable output paths	failure → terminal failed run; non-run → non-complete	executed or failed ; never reproduced by itself
3. Close	all expected cells terminal	expected/observed cardinality; target, order, time, spatial and mask hashes; prediction hashes; provenance links	incomplete or mismatched family → repair required	closed-unverified
4. Audit	closed matrix; metrics still locked	identity audit, boundary audit, control validity, repair parent/child hashes, prediction-change flag	any defect → invalidate affected formal family; non-selective rerun where selection could bias repair	audit-pass
5. Unlock	every registered audit passes	unlock authority, conditions, receipt hash, access counters, exact audited matrix hash	absent receipt or premature access → BLOCKED	metric-unlocked for the named role only
6. Decide	metric implementation and reference fixed	endpoint, reference, effect, uncertainty, consistency checks, conjunctive gate outcome	any required check fails → versioned hypothesis terminates	gate-pass or gate-fail ; subchecks stay typed
7. Write	decision and evidence chain retained	claim ID, authoritative artifact and hash, evaluation type, allowed wording, prohibited wording, scope limit	missing lineage → claim withheld	scoped manuscript statement

Table 8: Fail-closed audit steps (continued): wording rules that keep diagnostic, proxy, engineering, and procedural results separate from stronger claims.

Evidence type	Required wording	Example permitted wording	Unsupported wording
Oracle diagnostic	[target-dependent role] + [dataset boundary] + [reference] + [opportunity bound]	“indicates an infeasible upper bound”	“is deployable” or “can be recovered”
Proxy prediction	[proxy label] + [forecast-origin information set] + [discrimination/calibration]	“predicts relative expert advantage”	“improves forecast RMSE” or “identifies a cause”
Real-GT development/internal	[data role] + [reference] + [endpoint] + [effect/interval]	“improves internal rolling RMSE by $x\%$ ”	“generalizes”, “blind test”, or “state of the art”
Engineering measurement	[device + workload + precision + batch] + [endpoint] + [comparison]	“measured P50 was x ms on the frozen workload”	“is efficient” without a matching serving study
Conjunctive gate	[candidate] + [all required checks] + [pass/fail] + [authorized next action]	“failed the 1% point-effect requirement; stage stopped”	replacing the frozen question with a positive subcheck
Artifact-integrity event	[defect] + [pre-metric state] + [repair/audit] + [unlock outcome]	“blocked interpretation until identity audit passed”	“repair improved accuracy”
Procedural data boundary	[asset role] + [storage state] + [analytical-use state] + [threat-model limit]	“vault-materialized; analytical access after lock was zero”	“never materialized” or “cryptographically inaccessible”

Table 9: Model identity, local adaptation, and claim scope. “Unknown/not claimed” is used when the frozen project artifacts do not establish a unique upstream identity or an authoritative field.

Paper label	Frozen implementation identity	Data/input contract and local changes	Role and explicit non-claim
UrbanEV CAPER	Existing exact-target expert predictions; legacy artifact label phase_only . The latency artifact reports 47,992 parameters. Upstream paper/repository checkpoint identity: unknown/not claimed .	UrbanEV panel, 168-hour project context and four horizons in the shared evaluation. The paper reuses predictions and does not reconstruct an unrecorded upstream training recipe or covariate contract.	Single expert, fixed-fusion component, router expert, and KD anchor. Not claimed as an exact replay of an external CAPER release or as a generally defined architecture.
Compact TimeXer-168	Local compact TimeXer-family artifact; 2,358,788 parameters in the latency serving artifact. Exact upstream repository revision for the UrbanEV artifact: unknown/not claimed .	Existing UrbanEV predictions with 168-hour context and shared target identity. No new expert training was performed for the audit chain.	Best evaluated single UrbanEV model within the frozen candidate set. Not “official upstream TimeXer”, universal best, or SOTA.
Global Fixed	Deterministic convex combination $\alpha_f \hat{y}^T + (1 - \alpha_f) \hat{y}^C$; one fold-level scalar pooled across horizons. Serving count 2,406,781 parameters includes both experts and scalar.	Fold 1 uses paired pre-test validation; folds 2–6 use earlier-fold OOF predictions only. Current-fold targets cannot fit α_f .	Selected fixed reference and distillation teacher. Not a cost-free ensemble, blind-test result, or cross-dataset claim.
Router v1	Local CAPER_TimeXer_ReliabilityGate; direct-loss MLP has hidden widths 32/16, horizon embedding 4, dropout 0.1, Huber loss, seed 42. Authoritative deployed parameter count: unknown/not claimed .	Expert outputs/disagreement plus 21 forecast-origin state features, rolling cross-fit; no feature timestamp exceeds the forecast origin.	Frozen learned-router comparison. It failed the 1% gate; no robustness or deployment qualification is claimed.
Chronos-2	Official <code>amazon/chronos-2</code> snapshot resolved to commit <code>29ec3766d36d...e8498c</code> ; <code>chronos-forecasting 2.2.2</code> , Torch 2.5.1+cu121. Parameter count: unknown/not claimed .	Zero-shot FP32; 168-hour multivariate context; native Q0.5 output; no future covariates; predictions clipped to $[0, 1]$ for the frozen primary metric.	One teacher-qualification screen. Its failure is not evidence that foundation models generally fail for EV forecasting.
Residual student / aligned KD	Local CAPER-anchor plus shared DLinear residual branch; 338 trainable parameters. GT-only, aligned-KD, and shuffled-teacher variants share architecture, initialization, optimizer, data, epochs, and seed.	Occupancy-rate-only input, 168-hour lookback, 275 channels; aligned loss adds the frozen teacher-residual term with $\lambda = 0.5$.	Negative-gated distillation experiment. The aligned-vs-shuffled subcheck is not a passed improvement over GT-only and does not authorize latency or architecture search.
Paris CAPER	Paris_CAPER_phase_only , newly trained local adaptation; no UrbanEV checkpoint or fitted coefficient loaded. Authoritative parameter count: unknown/not claimed .	Development-only. Available, Charging, Passive, and Other map to available/active/unavailable; daily/weekly ordinal lags; target-history encoder plus station embedding replaces unavailable UrbanEV channels.	Paris single-model candidate. Must not be named unqualified CAPER or treated as an exact UrbanEV replay.
Paris TimeXer	TimeXer_local_audited_compact_L168 ; local source SHA-256 begins <code>563d4ce4304f</code> ; model class source matches the workspace copy. Upstream commit is unknown/not claimed .	Retrained from initialization on Paris development; compact profile ($d = 128$, four heads, two layers); external train-only scaler, masked loss, direct endpoint, no time markers.	Paris single-model candidate. Explicitly a local audited adaptation, not an exact official upstream implementation.
Paris strict teacher	Fold-specific convex combination of the two Paris local adaptations; no separately trained joint model and no protected-target input.	Fold 1 α from pre-evaluation validation; later folds from prior-fold OOF predictions; shared station, mask, target, and timestamp identities.	Development-only teacher qualification. It failed the point-effect gate; student training and formal/protected evaluation were not authorized.

Table 10: Research-question map from operational definition to bounded answer.

RQ	Operational question	Evidence and analysis	Direct result	Bounded answer
RQ1	How many frozen configurations are exact, disclosure-limited, trend-only, underidentified, failed, or non-complete?	1,224-row status registry; deterministic S1–S6 classifier; execution state, source-cell mapping, fold coverage, and worst disclosed-metric grade.	865 artifacts complete; S1=156, S2=216, S3=122, S4=371, S5=19, S6=340.	Completion is broader than exact reproduction in this inventory. Counts are not a field-wide reproducibility estimate or a model ranking.
RQ2	How much oracle complementarity survives feasible prequential fusion and learned routing?	Target-dependent oracle; expert residual diagnostics; forecast-origin proxy classifier; strict Global/Horizon Fixed controls; direct-loss and secondary router variants.	Oracle +16.926% versus TimeXer; Global Fixed +3.178%; primary router +0.443% versus Global Fixed.	Global Fixed retains a 3.178% internal point gain. The smaller router gain fails the frozen effect gate.
RQ3	What changes after synchronous latency and the minimum-effect rule are enforced?	RTX 4060 FP32 batch 1/8/32 benchmark; batch-1 primary endpoint; router, Chronos-2, KD, and Paris conjunctive gates.	Global Fixed has +48.86% batch-1 P50 versus TimeXer; later gains are +0.443%, −3.048%, +0.119%, and +0.847%.	The accuracy leader has a measured serving cost; positive or consistent subchecks do not automatically qualify the next engineering stage.
RQ4	How do artifact checks, separate storage, and stopping rules affect which claims are supported?	Metric lock; non-selective control rerun; 32/32 Paris prediction closure; unchanged-prediction repair; versioned termination; procedural vault guard.	Three identity shuffles were excluded before metrics; 10,236 masked target entries were restored without prediction changes; analytical access after the Paris lock remained zero.	The checks blocked invalid results and stopped failed versions. They do not prove higher accuracy, hardened security, or external validity.

A Reproducibility and Audit Appendix

This appendix documents the audit checks used in the case study. It reuses frozen main-workspace artifacts and performs no model training or inference. It does not recompute any experiment metric, run a new bootstrap or threshold sensitivity, or read Paris formal or protected target content. The tables separate facts stored in the artifacts from suggestions that may be useful elsewhere. When a field cannot be established from a frozen artifact, it is reported as *unknown/not claimed* rather than filled in by assumption.

The appendix does not add analyses that were not run. There was no independent-coder classification exercise, inter-rater agreement statistic, alternative bootstrap block-length analysis, or new Global Fixed uncertainty interval. The 0.5/1/2% comparison is arithmetic context only; it does not recompute metrics or change a decision. The S1–S6 registry is a deterministic single-pipeline classification of one inventory. Historical project records used a review label for what was actually a separately instantiated automated differential artifact audit, not a blinded external human replication or an inter-rater study.

A.1 Deterministic S1–S6 decision tree

The classifier in `experiments/00_registry_reclassification/scripts/rebuild_status_registry.py` applies the following ordered rules to each frozen manifest row. Order is part of the definition; a later numerical rule cannot overwrite an earlier execution fact.

- D1.** If `execution_status=failed`, assign S5 and require any future attempt to use a new versioned retry identifier.
- D2.** Otherwise, if `execution_status` is not `complete`, assign S6 and require any future execution to use a new versioned run identifier.
- D3.** Otherwise, if the row maps to UrbanEV source Table 4, assign S4 because the source paper and repository do not identify the disclosed node uniquely.
- D4.** Otherwise, if no source comparison cell can be mapped uniquely, assign S4.
- D5.** Otherwise, collect the disclosed RMSE, MAPE, RAE, and MAE grades and take the worst valid grade. The frozen bands are exact (relative error no greater than 2%), near (2–5%), and trend (greater than 5%); an empty valid-grade set defaults to trend when a comparison record exists.
- D6.** If comparison fold coverage is below six, assign S3. Otherwise, if the worst grade is trend, assign S3.
- D7.** Otherwise, if the track is `official` and the worst grade is exact, assign S1. Every other completed, uniquely mapped, non-trend row is S2.

The rules therefore apply in this order: S5/S6 → S4 → S1/S2/S3. A completion mark indicates process completion only; it is not a synonym for S1. The following examples are the first row of each final status in the authoritative CSV. The `classification_reason` text is an English rendering of the Chinese source field so that the appendix compiles under the paper’s Latin-font setup.

Table 11: One frozen registry example per evidence status. Full run identifiers and reasons are retained to make the classification path inspectable.

State	Example ID	Reason
S1	<code>official-compat-v1-t3-lo-occ-3-1-region-None-42-20-eae54b2d</code>	Official-compatible protocol is identifiable, all six folds are complete, and the relative error of each of four disclosed metrics is no greater than 2%.
S2	<code>audited-t3-lo-occ-3-1-region-None-42-20-efa1ea0d</code>	Six folds and the main protocol align, but an audited correction is present or at least one metric lies in the 2–5% approximation band.
S3	<code>audited-t3-ar-occ-3-1-region-None-42-20-3b0793b0</code>	At least one paper-disclosed metric has relative error above 5%; only trend-level evidence is retained.
S4	<code>audited-t4-lo-occ-3-1-194-None-42-20-42da10db</code>	Neither the paper nor repository discloses the Table 4 node identity; node 194 is a predeclared reverse-identification candidate, not a unique truth.
S5	<code>audited-t3-lstm-occ-6-3-region-None-42-20-c735895e</code>	The frozen configuration was executed but failed; the failure artifact is retained and cannot contribute a paper number.
S6	<code>audited-t3-TimesNet-occ-3-3-region-None-42-1-1687eb9d</code>	The configuration is frozen in the manifest but was not executed or registered after accumulated failures in the same model batch.

A.2 Model identity is a claim field

A model name alone does not identify the exact implementation. The same family label may refer to an upstream release, a local source snapshot, an existing prediction artifact, a retrained adaptation, or a fixed combination of other artifacts. Table 9 reports the identity recorded in the frozen artifacts and lists fields that remain unknown. This is especially important for Paris: `Paris_CAPER_phase_only` and `TimeXer_local_audited_compact_L168` are development-only local adaptations, not exact transfers of the UrbanEV experts or claims about every upstream implementation.

A.3 Fail-closed state machine

The audit states are:

`REGISTERED` $\rightarrow$ `RUNNING` $\rightarrow$ `CLOSED_UNVERIFIED` $\rightarrow$ `AUDIT_PASS` $\rightarrow$ `METRIC_UNLOCKED` $\rightarrow$
`GATE_PASS` or `GATE_FAIL`.

Three rules prevent a defective artifact from moving forward. First, an incomplete matrix, target/order mismatch, invalid control, boundary violation, or provenance failure moves `CLOSED_UNVERIFIED` to `BLOCKED` or `REPAIR_REQUIRED`; aggregate metrics remain inaccessible. Second, a permitted repair creates `REPAIRED_UNVERIFIED`, records parent and child hashes and whether predictions changed, and returns to the full audit. It does not move directly to metric unlock. If selective repair could use information about which controls failed, the full control family is invalidated and rerun. Third, `GATE_FAIL` is terminal for the current protocol version. Continuing with a changed model, threshold, loss, or evidence role requires a new version and a still-valid data boundary.

Two incidents show how these transitions worked. In M7-C, the first audit detected three identity permutations at 69/72 cells while `metrics_unlocked=false`. All 24 shuffled controls were invalidated and rerun before the 72/72 audit passed. In Paris, 32 prediction jobs closed without evaluation metrics. Missing-target representation was repaired in 24 artifacts (10,236 entries) without changing predictions, and development metrics were unlocked only after the 32/32 prediction audit passed. These are integrity events, not accuracy improvements.

A.4 Common lock schema and invariants

The project used several lock and receipt formats rather than one universal schema. Table 12 gives a common interface for the rules that connect each claim to its evidence. It does not imply

that every historical artifact contains every field. A missing field must be recorded as **unknown** or fail the relevant gate; it must not be inferred from a filename.

Table 12: Normalized lock schema. Existing examples use different field names; the rightmost column states the rule an implementation must enforce.

Group	Required fields	Existing artifact exemplar	Invariant
Protocol	<code>protocol_id</code> , version, status, protocol SHA-256	router, latency, distillation, and Paris teacher protocol files	Data role, endpoint, reference, gate, and stop rule are immutable before aggregate access.
Configuration	<code>run_id</code> , config fingerprint, model identity, seed, fold, horizon	status registry and run manifests	One identifier maps to one configuration; retries retain parentage rather than overwriting identity.
Impl.	code/source hashes; dependency and bundle hashes	Chronos model lock; Paris implementation fingerprints	The implementation that emitted predictions equals the registered implementation or has an explicit non-training amendment receipt.
Environment	Python and library versions; accelerator, runtime, precision; optional environment hash	latency hardware/software manifests; Chronos environment block	Scope is explicit. If no unified environment hash exists, reproducibility is limited rather than implied.
Data boundary	source/data hash, role, permitted metadata, fold dates	baseline protocol; Paris boundary lock and source manifest	Development, formal, and protected roles cannot be silently exchanged.
Target identity	target-value, index/order, time, spatial-unit, and mask hashes; fold-horizon cell	expert diagnostics, run status, prediction audits	Candidate and reference use identical target positions and semantics for every compared cell.
Prediction	prediction SHA-256, shape, finiteness/range, attempt path	per-run receipts and prediction audits	The audited bytes are the bytes aggregated; a repair changing predictions requires new scientific evidence.
Metric	metric script/hash, endpoint definition, output hash	metric-lock scripts, aggregate summaries	The metric implementation and weighting rule are fixed; output cannot precede unlock.
Unlock	lock time, conditions, authority, receipt SHA-256, matrix/state hash	Paris metrics-unlock receipt; M7-C formal audit	Every condition is conjunctive and machine-checkable; role-specific unlock does not unlock another role.
Access	analytical-access counter, operation definition, log/receipt location	Paris protected guard and prediction audit	Counter semantics and time origin are stated; storage migration is recorded separately.
Repair	parent hash, child hash, reason, prediction-changed flag, audit return state	shuffled-control amendment and masked-target repair ledger	Repair is traceable and returns to unverified state; non-selective scope is used where observation could bias repair.
Decision	reference, point effect, interval, fold/cell/horizon checks, gate vector, authorized next action	router, Chronos, D1, and Paris teacher summaries/decisions	A failed required component yields terminal failure for that version even when another subcheck passes.
Claim	claim ID, authoritative artifact/hash, evaluation type, allowed/prohibited wording, scope limit	Table 13	Manuscript wording cannot exceed the evidence type or data role.

A.5 Operational definition of analytical access

The Paris guards distinguish three operations.

Storage materialization. Copying, moving, splitting, preserving, or hashing target-bearing files without using their values. Creating or checking a path/size/hash manifest also belongs to this category. These operations are recorded as storage events and do not count as target analysis.

Approved metadata handling. Reading only the registered non-target fields (date boundary, station identifier, latitude, longitude, postcode, and area), row counts, file sizes, and hashes needed to establish the split and storage receipt. State and target values are not used.

Analytical access. Any post-lock use of formal or protected target values. This includes displaying, summarizing, plotting, filtering by, learning from, scoring against, or making a scientific claim from those values. Manual viewing also counts. Use in training, validation, model selection, metrics, threshold selection, error analysis, or a paper table counts even when no file is modified.

Under this definition, `analytical_access_count=0` means that no protected label was used after the boundary lock for the listed analytical operations. It does not mean that target bytes were never materialized: one storage migration created vault shards and moved recoverable sources. It also does not erase the logged display of two example rows during pre-lock schema discovery. The counter is supported by project receipts and audits. It is not enforced by the operating system, does not provide encryption, and cannot prevent deliberate bypass by an authorized user.

These controls are intended to prevent accidental use, or use chosen after earlier results were seen, within the project workflow. Examples include a script path that reaches a future role, a stale cache, recoverable Git objects, or a metric opened before the registered closure. The vault path, manifest, hashes, guards, and negative main-project scan reduce these risks. Operating-system compromise, deliberate bypass, encryption, and immutable access logging are outside scope. The supported claim is source separation and zero recorded analytical use, not hardened security.

A.6 Paris fold-synchronous bootstrap, including the partial block

The frozen implementation in `experiments/09_distill_v2/scripts/aggregate_paris_teacher.py` used 5,000 replicates and seed 42. For each development fold, it found target timestamps shared by all four horizons and sorted them. It required one contiguous hourly sequence and divided that sequence at offsets 0, 24, 48, and so on. The final slice was kept even when it contained fewer than 24 hours.

Each replicate sampled the fold’s original number of blocks with replacement. The same sampled timestamp sequence was used for every horizon in that fold. Observed-target masks were then applied within each cell, and cell RMSEs were macro-averaged. The better CAPER/TimeXer reference was selected again inside the replicate, and teacher gain was computed against that reference. The reported interval uses the 2.5th and 97.5th percentiles.

August and October each contain 732 common hourly timestamps: 30 full 24-hour blocks and one 12-hour block (31 selectable blocks). September and November each contain 708 timestamps: 29 full blocks and one 12-hour block (30 selectable blocks). Every block, including the 12-hour partial block, had the same probability of selection. Because the partial block contained fewer timestamps, replicate length varied with the number of times it was drawn. No padding, wraparound, discarding, or extra weighting was applied. The reported Paris interval applies to this exact rule. No alternative block length or partial-block sensitivity analysis was run.

A.7 Claim-to-artifact links

Table 13 links each headline result to the machine-readable artifact that supports it. The hash is computed over the named artifact in the main workspace. A hash identifies file bytes; it does not prove that the method is correct. The evaluation-type and wording columns state the claim limit. For example, the Global Fixed row supports an internal point comparison and a separate measured latency result. It does not gain an unreported bootstrap interval merely because later stages used one.

Table 13: Claim-to-artifact trace and wording guardrails (part 1 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256 ₁₂	Evaluation type	Allowed wording; prohibited wording
C1	Frozen registry: 1,224 configurations; 865 complete artifacts; S1=156.	experiments/00_registry_reclassification/outputs/CONFIG_STATUS_SUMMARY.json 833352c136a4	registry audit	<i>Allowed:</i> In this inventory, completion and exact reproduction differ. <i>Prohibited:</i> A field-wide reproducibility rate or a model-quality ranking.
C2	Source-table status distributions are uneven under the frozen mapping.	experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv 4b61c2aea510	registry audit	<i>Allowed:</i> Report S1–S6 counts by UrbanEV source table. <i>Prohibited:</i> Call an S4 model invalid or intrinsically weak.
C3	Target-dependent CAPER–TimeXer oracle gain is 16.926%.	experiments/01_expert_diagnostics/outputs/EXPERT_DIAGNOSTIC_SUMMARY.json 53fa7f22fca6	real-GT oracle diagnostic	<i>Allowed:</i> An infeasible upper bound indicates local complementarity. <i>Prohibited:</i> The oracle is deployable or its gain is recoverable.
C4	State-feature expert-win AUC is 0.6151.	experiments/01_expert_diagnostics/outputs/STATE_FEATURE_DECISION.json 9f84ac483e02	self-supervised proxy	<i>Allowed:</i> Relative expert advantage is moderately predictable. <i>Prohibited:</i> AUC proves lower forecast RMSE or causal demand knowledge.
C5	Strict Global Fixed RMSE is 0.071367, a 3.178% internal gain.	experiments/03_fixed_fusions/outputs/FIXED_FUSION_SUMMARY.json 0a0162204c83	internal rolling real GT	<i>Allowed:</i> Improves on TimeXer under the audited UrbanEV protocol. <i>Prohibited:</i> Blind-test, cross-city, efficient, or deployment-ready.
C6	Direct-loss router gains 0.443% and fails the frozen 1% gate.	experiments/05_router_v1/outputs/ROUTER_V1_SUMMARY.json a9dfc2ebe4a6	internal real GT; gated	<i>Allowed:</i> A positive sub-threshold internal signal; current stage stopped. <i>Prohibited:</i> Router qualification, robustness, or post-hoc gate revision.
C7	Global Fixed batch-1 P50 is 48.86% above TimeXer.	experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv f94d0a7a9e95	hardware-specific latency	<i>Allowed:</i> Measured on the frozen RTX 4060 FP32 sequential workload. <i>Prohibited:</i> Portable latency, low cost, or universal serving efficiency.

Table 14: Claim-to-artifact trace and wording guardrails (part 2 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256 ₁₂	Evaluation type	Allowed wording; prohibited wording
C8	Chronos-2 gains -3.048% versus Global Fixed and fails qualification.	experiments/06_5_teacher_qualification/outputs/CHRONOS2_QUALIFICATION_SUMMARY.json 01e8084c16ae	zero-shot internal screen	<i>Allowed:</i> This locked revision does not qualify for this protocol. <i>Prohibited:</i> Foundation models are generally inferior for EV data.
C9	Aligned KD gains 0.119% versus identical GT-only; CI crosses zero.	experiments/07_residual_distillation/outputs/D1_SUMMARY.json c38163157c89	internal real GT; gated	<i>Allowed:</i> The primary distillation comparison fails; distill_v1 stops. <i>Prohibited:</i> Distillation improves the student or merits D2 latency.
C10	Aligned KD gains 0.259% versus shuffled teacher.	experiments/07_residual_distillation/outputs/D1_SUMMARY.json c38163157c89	negative control	<i>Allowed:</i> Evidence of a small sample-aligned teacher signal. <i>Prohibited:</i> Engineering value or a passed primary KD claim.
C11	Paris teacher gains 0.847%, wins 4/4 folds and 16/16 cells, but fails 1%.	experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json 5b7c1ddc353c	development real GT	<i>Allowed:</i> Consistent development gain below the frozen qualification effect. <i>Prohibited:</i> Round to 1%, imply formal/protected performance, or call it official TimeXer.
C12	Three identity shuffles were blocked at 69/72 before metric unlock.	experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json 584313534390	artifact integrity	<i>Allowed:</i> The lock forced a non-selective rerun of all 24 controls. <i>Prohibited:</i> Use the invalid controls as performance evidence.
C13	Paris prediction audit passed 32/32 before development-metric unlock.	experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json ee882d8e907b	artifact integrity	<i>Allowed:</i> Identity checks authorized development-metric aggregation. <i>Prohibited:</i> The audit proves model correctness or improved RMSE.
C14	Protected analytical-access counter is 0; vault materialization is true.	experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json a7780e39f4ef	procedural data boundary	<i>Allowed:</i> No protected label was used analytically after the boundary lock. <i>Prohibited:</i> Targets never existed, were never materialized, or are cryptographically inaccessible.

A.8 Latency across the frozen batch grid

The main text uses batch-one end-to-end P50 as its primary engineering endpoint. Table 15 lists the complete batch 1/8/32 grid from the frozen latency summary. The script `experiments/10_audit_paper/scripts/generate_revision_appendix.py` checks all 18 method–batch rows and formats the stored values without rerunning the benchmark.

The hardware and software settings were Windows 10 build 26200, 16 physical/32 logical AMD CPU cores, approximately 16.8 GB RAM, an RTX 4060 Laptop GPU with driver 595.71, Python 3.10.8, Torch 2.5.1+cu121, CUDA 12.1, cuDNN 9.1, FP32, and single-threaded Torch/OMP/MKL host settings. GPU clocks were observed per process rather than pinned. Model loading and checkpoint I/O were excluded. Cold-start, quantized, compiled, parallel-expert, and serverless settings were not tested.

Table 15: Frozen end-to-end synchronous latency at batch sizes 1, 8, and 32. P50 and P95 are milliseconds per request; throughput is forecast origins per second at P50. Every value is formatted directly from `LATENCY_SUMMARY_TABLE.csv`; no benchmark was rerun.

Method	Batch origins	E2E P50 (ms)	E2E P95 (ms)	Origins s^{-1} (P50)
DLinear	1	1.166	1.763	857.9
DLinear	8	2.118	2.874	3776.9
DLinear	32	5.382	7.728	5945.7
NLinear	1	0.674	1.056	1483.7
NLinear	8	1.554	2.101	5148.0
NLinear	32	3.896	5.373	8213.7
GBDT-Forecaster	1	6.242	7.195	160.2
GBDT-Forecaster	8	40.188	50.261	199.1
GBDT-Forecaster	32	157.087	186.205	203.7
CAPER	1	7.934	10.798	126.0
CAPER	8	10.740	14.253	744.9
CAPER	32	19.753	24.266	1620.0
TimeXer	1	14.067	15.952	71.1
TimeXer	8	105.780	113.816	75.6
TimeXer	32	482.165	519.827	66.4
Global Fixed CAPER+TimeXer	1	20.940	33.123	47.8
Global Fixed CAPER+TimeXer	8	125.350	136.947	63.8
Global Fixed CAPER+TimeXer	32	479.634	510.574	66.7

Workload: one request produces all four horizons for 275 UrbanEV zones. Measurements used FP32, one RTX 4060 Laptop GPU, sequential expert execution, five processes, 50 warm-ups, and 1,000 timed iterations per process and method–batch cell. Throughput is not monotone across model families because the frozen serving graphs batch differently; these values should not be assumed to hold in other deployment settings.

A.9 Protocol and environment artifact index

The following index lists existing controls. Listing a file does not make missing environment information complete. The latency and Chronos stages have explicit environment records. Other stages retain protocols, source/configuration fingerprints, and run receipts but do not provide one cross-stage environment hash. Exact environment equivalence is therefore unknown/not claimed where that record is absent.

Table 16: Protocol and environment artifact index (part 1 of 2).

Artifact	Function	Scope limit
experiments/00_registry_reclassification/scripts/rebuild_status_registry.py experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv experiments/02_lightweight_baselines/configs/BASELINE_PROTOCOL.json refine-logs/ROUTER_PROTOCOL_V1_0.md and experiments/05_router_v1/configs/ROUTER_V1.json	Deterministic S1–S6 implementation and priority order Row-level run IDs, fingerprints, grades, reasons, and rerun policy UrbanEV target, folds, horizons, seed, and candidate protocol Router feature/training configuration and frozen Gate 4	Single programmed classifier; no inter-rater validation One frozen UrbanEV inventory Candidate-set internal evaluation only One seed; proxy and real-GT roles remain separate
experiments/06_static_latency/LATENCY_PROTOCOL_V1_0.md	Request shape, synchronization, warm-up, repetitions, and primary endpoint	Warm synchronous serving, not cold start
experiments/06_static_latency/hardware_manifest.json and experiments/06_static_latency/software_manifest.json	Device, driver, platform, runtime, libraries, precision, host threads	One laptop system; clocks observed rather than pinned
experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv	Batch 1/8/32 model-only and end-to-end distributions, throughput, memory	Hardware/workload-specific
experiments/06_5_teacher_qualification/TEACHER_QUALIFICATION_PROTOCOL_V1_0.md experiments/06_5_teacher_qualification/CHRONOS2_MODEL_LOCK.json	Chronos-2 qualification endpoint and gate Resolved model commit, file hashes, snapshot bytes, and environment	Exact project protocol only One revision and zero-shot profile
experiments/07_residual_distillation/DISTILL_PROTOCOL_V1_0.md experiments/07_residual_distillation/configs/FORMAL_STUDENT_CONFIG_V1.json	Teacher, student, loss, controls, milestones, stop rules 338-parameter student input/architecture/training identity	Distill v1 only; later search prohibited after failure One architecture and seed
experiments/07_residual_distillation/SHUFFLE_CONTROL_SPEC.json	24-hour block-shuffle negative-control construction	Control test, not a second primary endpoint

Table 17: Protocol and environment artifact index (part 2 of 2).

Artifact	Function	Scope limit
experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json and experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT.json	Pre-unlock failure and final 72/72 audit	Supports control integrity, not accuracy
experiments/07_residual_distillation/outputs/D1_SUMMARY.json and experiments/07_residual_distillation/outputs/D1_DECISION.md	Primary/negative-control effects and terminal gate decision	No D2 latency, multi-seed, or protected evidence
experiments/08_paris_new_evidence/DATA_BOUNDRY_LOCK.json and experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json	Role dates, permitted metadata, storage state, access counter, unlock rule	Procedural guard; not hardened authorization
experiments/08_paris_new_evidence/PARIS_EVIDENCE_VAULT_REPORT.md	Vault identities, hashes, negative main-project scan, threat limits	Metadata/hash audit only; no content-level protected audit
experiments/09_distill_v2/PARIS_TEACHER_QUALIFICATION_PROTOCOL_V1_0.md	Development folds, target, missingness, time rule, teacher fit, gate	Paris development only
experiments/09_distill_v2/PARIS_TEACHER_MODEL_CONFIG.json	Local CAPER/TimeXer profiles, training, queue, bootstrap	Local adaptations, not exact upstream implementations
experiments/09_distill_v2/TIMEXER_LOCAL_ADAPTATION_PROVENANCE.json and experiments/09_distill_v2/PARIS_CAPER_ADAPTATION_V1_0.md	Source hash and explicit local adaptation differences	Upstream TimeXer commit undisclosed; Paris CAPER newly trained
experiments/09_distill_v2/PARIS_TEACHER_DEPLOY_APPROVAL_V1_1.md	Automated differential audit closure after the earlier block	Automated artifact audit, not blinded human replication
experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json and experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_METRICS_UNLOCK_RECEIPT.json	32/32 closure, identities, role-specific metric unlock, access counts	Development-metric authorization only
experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json	Point effect, fold/cell results, synchronized bootstrap, gate vector	Current hypothesis failed; no student/formal/protected result
experiments/09_distill_v2/PARIS_TEACHER_RESULT_REVIEW.md	Separately instantiated result-to-claim check and provenance warnings	Warnings limit stronger process claims; they do not reverse the gate
experiments/10_audit_paper/scripts/generate_revision_appendix.py	Mechanical hash/latency table generation	No model, metric, resampling, or vault operation

A.10 Author and reviewer application checklist

Table 18 summarizes checks that authors and reviewers can apply. Missing validation must be stated as missing. The 1% rule is a local convention without an economic utility model. The S1–S6 labels come from one deterministic classifier and have no inter-rater evidence. The model and latency conclusions are limited to their recorded seeds, panels, adaptations, and serving setup.

Table 18: Author and reviewer checklist derived from this study.

ID	Check	Author must expose	Reviewer test
1	Target and role semantics	numerator/denominator, spatial unit, mask, sampling clock, and development/formal/protected role	Could two datasets or target constructions be mistaken for the same RMSE scale?
2	Configuration identity	run ID, protocol version, fingerprint, source mapping, seed, fold/horizon, and execution state	Could a completed run still be underidentified or map to several disclosed cells?
3	Information timing	forecast origin, lookback, covariates, fill/scaler fit interval, and prequential weight/tuning source	Does any current/future target influence a parameter evaluated in the same fold?
4	Artifact closure	expected matrix cardinality, terminal attempts, prediction/target/order/time/spatial/mask identities	Are aggregates withheld until every required cell is closed and verified?
5	Metric lock	metric implementation identity, lock state, unlock conditions, authority, receipt, and access counter	Can any result be read or selected before the audit passes?
6	Comparator and endpoint	primary reference, macro/micro weighting, endpoint, sign convention, and secondary diagnostics	Was the reference or endpoint changed after seeing results?
7	Uncertainty and gate	resampling unit, partial-block rule, replicates/seed, minimum effect, consistency checks, stop rule	Is a positive interval against zero being substituted for a failed effect-size requirement?
8	Controls and repair	negative-control construction, invalidation scope, before/after hashes, and whether predictions changed	Were only visibly bad cells repaired, or was the formal control family rerun non-selectively?
9	Model identity	upstream/local source, checkpoint/revision, local adaptations, input contract, and every unknown field	Does a local adaptation masquerade as an official or exact upstream implementation?
10	Engineering measurement	device, software, precision, batch/request shape, synchronization, warm-up, repetitions, P50/P95, throughput	Is a hardware-specific measurement being presented as general efficiency?
11	Future/protected evidence	storage versus analytical access, allowed metadata, threat-model limit, logs, and unlock rule	Does the boundary prevent analysis, merely record storage, or claim stronger security than implemented?
12	Claim source	claim ID, authoritative artifact, SHA-256, evaluation type, allowed/prohibited wording, scope limit	Can every headline number be traced, and are diagnostics/subchecks kept at their original evidence type?
13	Negative result	terminal decision, disallowed same-interval actions, and requirements for a new version/evidence boundary	Was a failed hypothesis silently continued through tuning, threshold changes, or a new structure?
14	Missing validation	limitations must name absent multi-seed, multi-hardware, external-team, utility, or rater evidence	Is any absent analysis implied by rhetoric such as robust, validated, secure, or general?

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